WATP

SEMESTER ONE

MATHEMATICS APPLICATIONS UNIT 1

2018

SOLUTIONS

Calculator–free Solutions

1.	(a)	SI = \$500 x 0.08 x 2	✓	
		= \$80	✓	
	(b)	$500 = 2000 \times \dfrac{R}{100} \times 4$	✓✓	
	(c)	(i) \$1000	✓	
		(ii) \$200	✓	
		(iii) \$1200	✓	[7]
2.	(a)	$M = (6)(2) + 5 = 17$	✓	
	(b)	$N = (6 + 2)^2$		
		$= 8^2 = 64$	✓	
	(c)	D4 = A4*B4+C4	✓✓	
	(d)	\$7.50	✓	
	(e)	GST = \$8	✓	
		Pre GST = \$80	✓	
	(f)	$\% = \dfrac{30}{120} \times 100$	✓	
		= 25%	✓	
	(g)	Score = 1.2 x 25	✓	
		= 30 goals	✓	
	(h)	$0.95x = 285$	✓	
		$\therefore \quad x = \dfrac{285}{0{\cdot}95} = 300$	✓	
		∴ 15 m was removed	✓	[14]
3.	(a)	$\dfrac{6}{10} = 0{\cdot}6$ and $\dfrac{12}{20} = 0{\cdot}6$	✓	
		Angles between two corresponding sides are congruent	✓	
		∴ Similar (SAS)	✓	
	(b)	$\dfrac{k}{12{\cdot}4} = \dfrac{12}{6}$	✓	
		$\therefore \quad k = (2)(12.4) = 24.8$	✓	
	(c)	(i) p + q = y + z	✓	
		(ii) 180	✓	
	(d)	Scale factor = 2		
		∴ Area = 4A	✓	[8]
4.	(a)	2 x 3	✓	
	(b)	(i) $\begin{bmatrix} 5 \\ 11 \end{bmatrix}$	✓✓	
		(ii) $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$	✓✓	
		(iii) $\begin{bmatrix} 2 & 4 & 6 \\ 8 & 10 & 12 \end{bmatrix}$	✓✓	
		(iv) $\begin{bmatrix} -3 & -6 \\ -9 & -12 \end{bmatrix}$	✓✓	[9]

5. (a) 60 cm ✓
(b) 9:16 ✓
(c) 27:64 ✓
(d) $4\pi(80)^2$ ✓
$= 25\ 600\pi\ \text{cm}^2$ ✓ [5]

6. (a)

$$\begin{array}{cc} & \text{To} \\ & \begin{array}{ccc} A & B & C \end{array} \\ \text{From} \begin{array}{c} A \\ B \\ C \end{array} & \begin{bmatrix} 0 & 1 & 2 \\ 2 & 0 & 2 \\ 1 & 1 & 0 \end{bmatrix} \end{array}$$

✓✓

(b) $\begin{bmatrix} 4 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 1 & 4 \end{bmatrix}$

✓✓

(c) Gives the two–stage routes between locations. ✓
(d) (i) 2 ✓
(ii) None. ✓ [7]

<u>Calculator–assumed Solutions</u>

7. (a) $\frac{19}{2}$ or 9·5 ✓
(b) 0 ✓
(c) $7 = \frac{3x + 1}{x - 4}$ ✓
$\therefore \quad x = 7.25$ ✓ [4]

8. (a) $\frac{8}{10} = \$0{\cdot}8$ US ✓
(b) $5 \times 33{\cdot}5 = 167{\cdot}5$ Riggit ✓
(c) $1000 + \frac{1\ 000\ 000}{105\ 000} \times 10 + \frac{2000}{33{\cdot}5} \times 10$ ✓
$= 1000 + 95.24 + 597.01$ ✓
$= \$1692.25$ ✓
(d) \$A597.01 = $597{\cdot}01 \times 10\ 500 = 6\ 268\ 656$ Rupiah ✓✓
(e) \$A30 = 30 x 10 500 = 315 000 Rupiah ✓
Total = 365 000 Rupiah $\therefore$ Generous ✓ [9]

9. (a) \$8500 ✓
(b) $\frac{8500}{30\ 000} \times 100 = 28{\cdot}33\%$ ✓✓
(c) $21\ 500 = 30\ 000(1 - r)^3$ ✓
$\therefore \quad r = 0.105$ ✓
$\therefore \quad 10.5\%$ ✓ [6]

10. (a) kilolitres ✓

(b) $\dfrac{12\cdot14}{63} = 19c$ ✓

(c) \$101.44 ✓

(d) \$153.71 ✓

(e) $0\cdot0048 \times 153\cdot71 = \$0\cdot74$ ✓

(f) $227 \times \dfrac{0\cdot1188}{365} \times 25$ ✓

= \$1.85 ✓ [7]

11. (a) (i) $x^2 = 17^2 + 15^2 = 514$ ✓

$\therefore x = 22.67$ ✓

(ii) $x^2 + 13^2 = 28^2 \rightarrow x^2 = 615$ ✓

$\therefore x = 24\cdot80$ ✓

(b) (i) $1\cdot2^2 + 1\cdot5^2 = 3\cdot69$ ✓

$1\cdot92^2 = 3\cdot69$ ✓

Since the sum of the squares of the sides = hypotenuse^2 ✓

Then right angled

(ii) Measure both diagonals.

If they are equal, then it is a rectangle. ✓✓ [9]

12. (a) Tax = (25 000 – 18 200)(0.19) = 1292 ✓

$\therefore \quad \dfrac{1292}{25\ 000} \times 100 = 5\cdot17\%$ ✓✓

(b) Taxable income = 22 000 – 4632 = \$17 368 ✓

Tax = 0 ✓

(c) \$37 000 ✓

(d) $600 = x(0\cdot19)$ ✓

$\therefore \quad x = \dfrac{600}{0\cdot19} = 3157\cdot89$ ✓

$\therefore$ Income = \$18 200 + 3157.89 = \$21 358 ✓ [9]

13. (a) Loan A: SI = \$6000 x 0.06 x 3 = 1080 ✓

Loan B: $A = 6000\left(1 + \dfrac{0\cdot055}{12}\right)^{36} = 7073\cdot69$ ✓✓

$\therefore$ CI = \$1073.69 ✓

$\therefore$ Loan B is better ✓

(b) $720 - P = P \times 0\cdot07 \times 2$ ✓ ✓

$\therefore \quad P = \$631\cdot58$ ✓ [8]

14. (a) $H = \begin{bmatrix} 3 & 2 & 5 \\ 6 & 3 & 8 \end{bmatrix}$ ✓

(b) $C = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$ ✓

(c) $HC = \begin{bmatrix} 32 \\ 53 \end{bmatrix}$ ✓

Cost of pack A is \$32, cost of pack B is \$53 ✓

(d) (i) $\begin{bmatrix} 30 & 40 \end{bmatrix}\begin{bmatrix} 3 & 2 & 5 \\ 6 & 3 & 8 \end{bmatrix}$

$= \begin{bmatrix} 330 & 180 & 470 \end{bmatrix}$ ✓

$\therefore$ 330 coriander, 180 basil and 470 cumin ✓

(ii) $\begin{bmatrix} 30 & 40 \end{bmatrix}\begin{bmatrix} 32 \\ 53 \end{bmatrix} = \begin{bmatrix} 3080 \end{bmatrix}$

$\therefore$ \$3080 ✓ [7]

15. (a) 19 x 8 x \$30 + 8 x \$30 x 1.5 = \$4560 + \$360 ✓

= \$4920 ✓

(b) $\frac{1}{1\cdot5} = \frac{2}{3} \rightarrow 8 \times \frac{2}{3} = 5\frac{1}{3}$ hours ✓

(c) 5040 = 18 x 8 x 30 + 2 x 1.5 x 8 x 30 ✓

$\therefore$ 2 public holidays ✓

(d) 1.05 x 30 = \$31.50 per hour ✓

(e) Increase of 5% per hour applies to all hours

$\therefore$ Increase of 5% ✓ ✓ [8]

16. (a) $AC^2 = 6^2 + 8^2 = 100 \rightarrow AC = 10$ ✓

$\therefore$ Area of Circle = $\pi \times 5^2 = 78\cdot54 \text{ cm}^2$ ✓

$\therefore$ Area of $\Delta ABC = \frac{1}{2} \times 6 \times 8 = 24\text{cm}^2$ ✓

Hence, Shaded region = 78.54 – 24 = 54.54 cm^2 ✓✓

(b) (i) $V = \pi \times 6^2 \times 15 = 1696\cdot46 \text{ cm}^3$ ✓✓

(ii) $SA = 2 \times \pi \times 6^2 + 2 \times \pi \times 6 \times 15$ ✓✓

$= 791.68 \text{ cm}^2$ ✓ [10]

17. (a) (i) B didn't bring A a present. ✓

(ii) A ✓

(iii) 2 ✓

(iv) A and C ✓

(b) (i) $\begin{bmatrix} 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ ✓✓

(ii) No meaning in this context ✓ [7]

18. (a)

$$\begin{array}{cc} & \begin{array}{ccc} & \text{To} & \\ A & B & C \end{array} \\ \text{From} \begin{array}{c} A \\ B \\ C \end{array} & \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 1 & 1 & 2 \end{bmatrix} \end{array}$$

✓✓

(b)

$$\begin{array}{cc} & \begin{array}{ccc} & \text{To (In 2 stages)} & \\ A & B & C \end{array} \\ \text{From} \begin{array}{c} A \\ B \\ C \end{array} & \begin{bmatrix} 4 & 4 & 0 \\ 0 & 4 & 0 \\ 4 & 5 & 4 \end{bmatrix} \end{array}$$

✓✓✓ [5]

19. Chocolate bars: 10 bars cost $\dfrac{520}{20} = 26$ c/bar = 2.6 c/g

✓ Chocolate slab: 1 slab costs $\dfrac{380}{150} = 2{\cdot}53$ c/g ✓

∴ Slab is better
buy ✓ [3]

20. (a) 2 cm x 200 = 400 cm ✓

∴ 4 m ✓

(b) Yes. ✓

They are similar triangles. ✓

(c) Real triangle has sides of 2 m, 3 m and 4 m. ✓

∴ $4^2 = 2^2 + 3^2 - 2(2)(3)\cos\theta$ ✓

∴ $\theta = 104{\cdot}5°$ ✓

∴ Area = $\dfrac{1}{2} \times 3 \times 2 \times \sin(104{\cdot}5)° = 2{\cdot}9 \text{ m}^2$ ✓ [8]